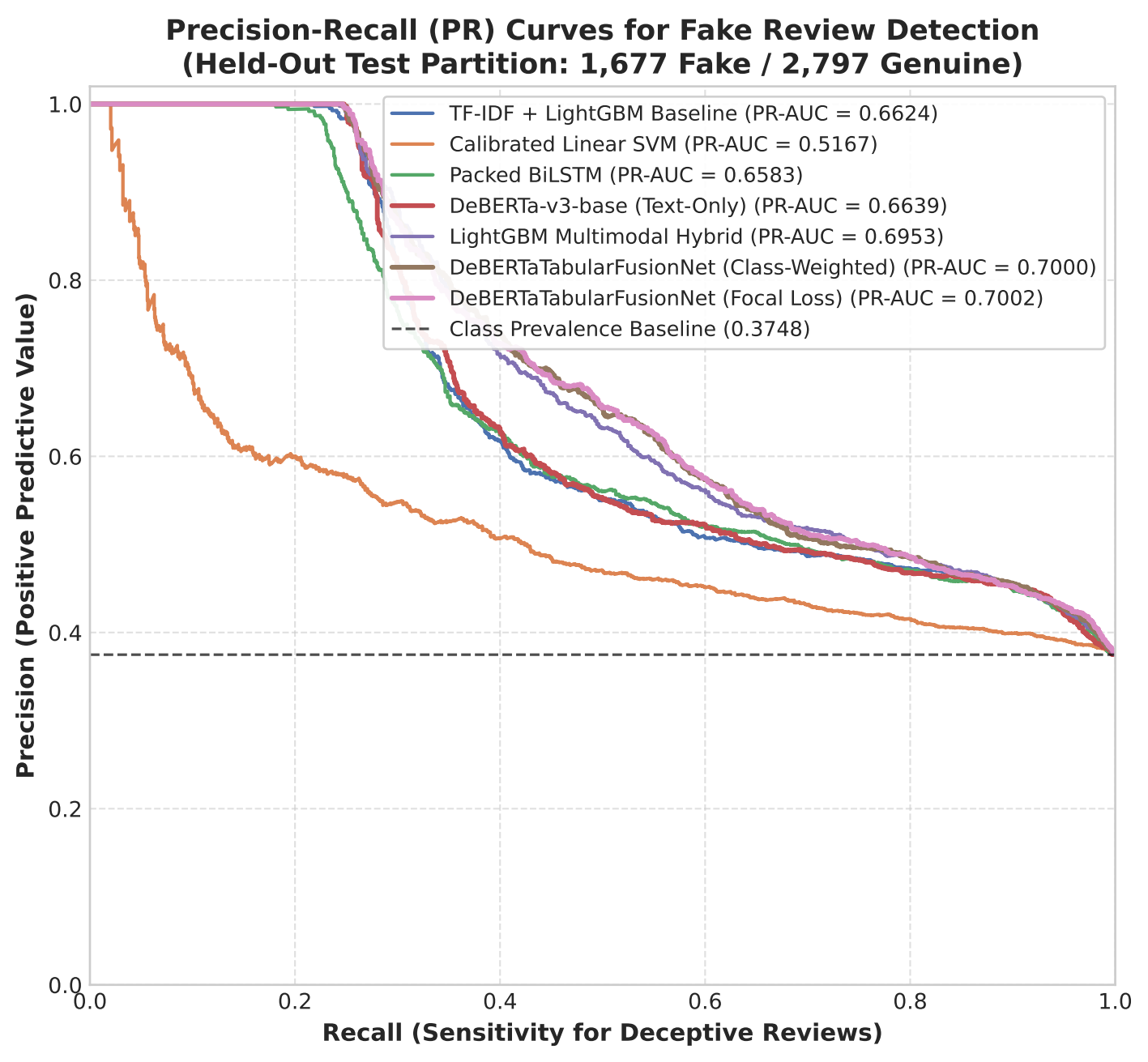

Precision-Recall (PR) Curves for Fake Review Detection
(Held-Out Test Partition: 1,677 Fake / 2,797 Genuine)
TF-IDF + LightGBM Baseline (PR-AUC = 0.6624)
Calibrated Linear SVM (PR-AUC = 0.5167)
Packed BiLSTM (PR-AUC = 0.6583)
DeBERTa-v3-base (Text-Only) (PR-AUC = 0.6639)
LightGBM Multimodal Hybrid (PR-AUC = 0.6953)
DeBERTaTabularFusionNet (Class-Weighted) (PR-AUC = 0.7000)
DeBERTaTabularFusionNet (Focal Loss) (PR-AUC = 0.7002)
Class Prevalence Baseline (0.3748)
Precision (Positive Predictive Value)
Recall (Sensitivity for Deceptive Reviews)
0.0
0.2
0.4
0.6
0.8
1.0